Track0tic™ · Device QR × Passport QR — how they interact

One rule: scanning a **device QR** binds Passport QRs to that device · scanning a **Passport QR** shows the record. Demo Tenant · pharma · 30 Oct 2026

Legend: Operator action · Server/device · Record / output · Exception

TRACK0TIC (issuer)

Device QR
printed once at assembling centre
PET label · TRK-0418

Passport QR batch
generated per tenant · opaque token
PP-AD-012001 → 013000

Tenant prints & sticks
PDF A4 or ZPL · PP cryo-adhesive
one QR per moving item

Both QRs hold only a token URL.
Everything shown comes from the server;
a damaged label is a reprint of the same token.

OPERATOR · phone (device QR scan page, mobile + OTP)

① Scan device QR
at warehouse
sign in · mobile + OTP
role = Operator

② Scan Passport QRs
1 … n, duplicates ignored
optional: going to, invoice
tap BIND

③ Travel · bag closed
multi-chemist route OK
bag open away from a stop
→ alert at office

④ Scan device QR at stop
"You are at a stop" (GPS)
scan only the QRs handed over
tap DELIVERED · rest stay bound

⑤ Next stop …
repeat ④ per chemist
leg ends when device is
empty or back at warehouse

HUB works the same: scan the cold-room HUB's device QR → bind = "in storage" leg · scan onto a Pro/Lite later = out of storage

DEVICE & SERVER (telemetry every 90 s → joined to Passport QRs by binding)

Server joins readings to Passport QRs via bindings — no firmware change needed.

BINDING created
passport ↔ device · time · user
previous binding auto-released

TELEMETRY
temp · RH · GPS · bag · weight
no passport in the payload

HANDOVER of record
for the QRs scanned at that stop
leg verdict computed per item

JOURNEY record
legs · stops · KM · halts
verdict per item · PDF

CHEMIST · PATIENT · PARTNER (Passport QR scan, no login)

A Passport QR never changes and never closes: whoever receives it just binds it to their own device QR.

Chemist with a device
scans own HUB/Pro device QR
→ binds received items
(optional, never a gate)
their leg starts

Chemist without a device
does nothing
Operator's Delivered scan is
the record; next step shows
"where monitored"

⑥ Patient scans Passport QR
phone camera · no login
English / हिन्दी
whole chain, anonymised
no tenant names · no TRK ids

⑦ Compliance PDF
last leg graph + earlier
legs as summaries
verification token
generation time shown

EXCEPTIONS (handled, never block)

Passport QR already bound elsewhere
bind anyway → previous binding released, previous custodian shown on the row

Wrong QR scanned
no unbind: scan it onto its correct device and it moves

Excursion on the leg
stays on every item's record and PDF; Delivered does not hide it

Held overnight at a stop
stop has HUB → bind to it;
no HUB → Pro stays with the goods

Label damaged · phone offline
reprint same token (logged) ·
type the id instead of scanning